



Pokhara University
Everest Engineering College

Final Internal Assessment
Spring- 2025

Level: Bachelor **F.M. 100**
Program: BE IT(6th Semester) **P.M. 45**
Faculty: Science & Technology **Time: 3hrs**
Section:
Subject: Internet Of Things

Attempt all the questions.

- 1 a) Define IoT with explanation of all characteristics of IoT. Also explain in detail about the basic components of an IoT system. 8
b) List the challenges of IoT. Explain the 4-layer architecture of IoT. 7
- 2 a) Describe with clean and neat diagram on how MQTT works. 8
b) What is CoAP? How can you use it to leverage it for IoT related applications. 7
- 3 a) What are the key characteristics of an actuator? Suppose you are to select an actuator for a controlling the shutter of your house, which kind of motor do you think you need and what characteristics would you prioritize for? 8
b) List all the datatypes available in Python programming language and write a python program to run on Raspberry pi which can blink an LED connected to GPIO17 pin of the raspberry Pi for 10 times a second. 7
- 4 a) Draw a clean block diagram of raspberry pi and describe its features. Also explain what kind of I/O are supported by the raspberry pi. 7
b) Suppose you are given the responsibility to implement a weather station that monitors the temperature, humidity and pressure in Everest College with battery powered option that charges with solar panel. What considerations should you put in mind regarding selection of microcontrollers, sensors and actuators with emphasis on platform, power efficiency, scalability and expansion. 8

- 5 a) Describe in detail about the SPI protocol and explain the modes of SPI supported by the raspberry pi with appropriate diagram. 8
- b) Provided is a DHT22 sensor which can read the temperature and humidity. Write a program to read the temperature and humidity off the sensor each 60 seconds and store it in a csv file. 7
- 6 a) What is matplotlib? What are the roles of cloud computing in IoT? Describe about emerging technologies in IoT and cloud computing. 7
- b) What are the best security practices that can be used to secure the IoT systems that we develop. What things needs to be put into consideration prior to designing the IoT system such that the security is maintained throughout the project. 8
- 7 Write short notes on: (Any two) 2*5=10
- a) Arduino
 - b) I2C protocol
 - c) Websockets

*****Best Wishes*****



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Final Internal Assessment
Spring- 2025

Level: Bachelor
Program: BE IT(6th Semester)
Faculty: Science & Technology
Section: A/B
Subject: Engineering Management

F.M. 100
P.M. 45
Time: 3 Hrs

Attempt all the questions.

- 1 a) What is management? Discuss about the various levels of management? Write down the characteristics of organization. 7
b) What are the roles and responsibilities of an Engineering Manager? What are the various skills necessary to be an efficient manager? 8
- 2 a) What is Planning? What are the various types of planning? Define them. Discuss about the limitation of planning. 8
b) What is organizing? Write down the various processes of organizing. Discuss about the merits and demerits of organizing? 7
- 3 a) What is motivation? What are the various techniques of motivation? Is fear a form of motivation? Discuss. 7
b) Describe the concept of leadership. What are its characteristics? What are the different styles of leadership? Explain. How does a leader differ from a boss? Distinguish. 8
- 4 a) What is human resource management? What are the objectives of HRM? Write down its function with relevant examples? 8
b) What is Job? Differentiate between Job Analysis Vs Job Specification Vs Job Description. What is the use of Job Analysis? 7

- 5 a) What is recruitment? What are the various sources of recruitment? Write about the various methods of recruitment.
- b) What is training? Write down the features and benefits of training. Differentiate between on job training Vs off job training.
- 6 a) What is quality management? Discuss about the principles of quality management. Write down the benefits of quality management. What is innovation management and disruption?
- b) What is conflict? Describe various levels of conflict. What is conflict resolution? What are the various methods you can adopt to settle or resolve a conflict?
- 7 Write short notes on: *(Any Two)* (5*2=10)
- a) Organizational Structure.
 - b) Performance Appraisal
 - c) SWOT Analysis
 - d) Maslow's Hierarchy of needs
 - e) ISO 9001

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Spring- 2025

Level: Bachelor

F.M. 100

Program: BE IT

P.M. 45

Faculty: Science & Technology

Time: 3hrs

Subject: Data Communication (6th Semester)

Attempt all the questions.

- 1 ☒ a) What is the role of standard organization in data communication? 7
☒ Differentiate between analog and digital data communication system.
- b) Differentiate between serial and parallel data transmission. Calculate 8
the channel capacity having bandwidth and SNR of 6 Hz and 6 dB respectively.
- 2 ☒ a) Check whether the given system is linear, casual and time invariant or 7
not.
i) $y(t) = \log x(t)$
ii) $y(t) = tx(t)$
- b) Define unit step and impulse signal with mathematical expression. 8
Find the energy and power signal $x(t) = e^{-2t} \cdot u(t)$ and justify whether it is energy signal or power signal.
Or
Define system. Describe basic system properties of signal with examples.
- 3 ☒ a) Define electromagnetic spectrum for telecommunication. Briefly 8
explain ground wave, sky wave and line of sight (LOS)
- b) Briefly explain TCP/IP model. How CSMA/CD detects and handles 7
collision?

- 4 a) Briefly explain attenuation, distortion and noise that occurs in data communication. Why are they called transmission impairments? 7
- b) Define attenuation, distortion. A bit stream is 10011101 is transmitted using the standard CRC method. The generator polynomial is x^2+1 . Derive the actual bit string transmitted. Show that the error is detected if any one bit in the received bit stream is inverted. 8

or

Encode the given five source symbols(messages) using Huffman codes and find its efficiency.

Message	M1	M2	M3	M4	M5
Probability	0.4	0.2	0.2	0.1	0.1

- 5 a) What are the data link layer design issue? How does Go-Back-N Automatic Repeat Request(ARQ) improve the Stop-and-Wait protocol. 7
- b) Define multiplexing and briefly explain its type. Compare among circuit switching, message switching and packet switching. 8
- 6 a) Differentiate between encoding and modulation? Explain differential PSK with example/ 7
- b) Encode the bit stream 1100101101101 using i) Unipolar ii) Polar RZ iii) Polar NRZ iv) AMI v) Manchester vi)HDB3 8

- 7 Write short notes on: (Any two)

2*5=10

- a) Standard Organization
- b) TCP/IP vs OSI
- c) VSAT

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Pokhara University
Everest Engineering College
Final Internal Assessment
SPRING-2025

Level: Bachelor

Program: BE IT

Faculty: Science & Technology

F.M. 100

P.M. 45

Time: 3hrs

Subject: Computer Network New (6th Semester)

Attempt all the questions.

1	a) Define Computer Network. Explain about Network Model. And discuss the criteria necessary for an effective and efficient network?	2+5
	b) Explain Protocol Stack, Interface and Services. Explain about protocol Hierarchies in detail?	4+4
2	a) What do you mean by Transmission Media? Differentiate circuit switching and packet switching.	3+4
	b) What is Forward Error Correction (FEC). What is the significance of hamming code in Error Detection, Explain with suitable Example.	8
3	a) why access control of channel is essential? Describe in detail about IEEE standard 802 for LANs.	8
	b) What are the design issues related to Transport layer? Explain TCP and its header segment.	7
4	a) Discuss about Interior and Exterior Routing protocols. What is count to infinity problem in distance vector.	7
	b) Pokhara university has 3 sub division located at Pokhara as head office, Kathmandu as Examination office and Biratnagar as contact office with 125,60 and 49 hosts respectively. Now you as network administrator design the network with below details. i) Calculate Broadcast address, Network address, usable address along with Subnet mask. ii) ISP provide IP address as 10.0.17.0/22	8

5	a) What do you mean by congestion? Explain traffic shaping in details. b) What is the role of URLs and Web Browsers in WWW. Explain about DNS.	7 8
6	a) What is NMS. Explain about various functions of network management. b) What is the differences between substitution Ciphers and Transposition Ciphers? Give example of each type.	7 8
7	Write short notes on: (Any two) a) ATM b) NGN c) Piggybacking	2*5 =10

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Pokhara University
Everest Engineering College
Final Internal Assessment
Spring- 2025

Level: Bachelor
Program: BE IT(6th Semester)
Faculty: Science & Technology
Section: A/B
Subject: Data Science and Analytics

F.M. 100
P.M. 45
Time: 3hrs

Attempt all the questions.

- 1 a) Why do we require data transformation? Explain by giving realistic examples.
- b) You are given thirty numbers: 3.22, 5.38, 4.95, 4.87, 6.67, 4.46, 4.55, 5.42, 5.85, 4.81, 6.5, 5.66, 4.32, 5.21, 6.07, 3.97, 5.3, 5.37, 5.17, 5.35, 4.47, 5.98, 5.93, 5.5, 6.73, 4.33, 6.8, 4.76, 5.37, 6.15. Do you think the distribution is uniform, exponential, or normal? Justify using histogram plot.
- 2 a) Define a power law distribution and give real-life examples for it. Why power laws are called heavy-tailed distribution? Explain by comparing power laws to normal and exponential distribution?
- b) Suppose you surveyed 5 male students and found their average daily spending on refreshments is Rs. 730, with a sample standard deviation of Rs. 200. A previous study shows that female students spend Rs. 810 per day on average. Using a t-test, determine if male students' daily spending is significantly different from that of female students. (Use $|t| > 1.8$ as the threshold for significance.)
- 3 a) How is the multivariate linear regression problem formulated in matrix notation? How is the solution obtained? Describe.
- b) You are given the following data.

x	0.5	1.0	1.5	2.0
y	-2.3	0.5	9.3	20.8

Estimate the value of y at $x = 1.3$ using the Nadaraya-Watson regression estimator. Take bandwidth $h = 0.2$

- 4 a) Explain Principal Component Analysis as a tool of identifying the latent variables in the data. 7
- b) Suppose that while building a decision tree classifier, the current node corresponds to the following data. 8

A	B	Class label
T	F	Black
T	T	Black
T	T	Black
T	F	White
T	T	Black
F	F	White
F	F	White
F	F	White
T	T	White
T	F	White

Should the decision tree algorithm use the attribute A or the attribute B for splitting? Justify. You can use any purity index, Gini or entropy, for the problem.

- 5 a) Perform one iteration of k-means clustering given the following data. 7

Items	x	y
A	5	3
B	-1	1
C	1	-2
D	-3	-2

Take the number of centroids $k=2$. Assume that initially A and B are allocated to cluster 1 and C and D are allocated to cluster 2.

- b) Why is logistic regression required? Describe how logistic regression models the probability $P(y = 1 | x)$ where y is a binary variable. 8

- 6 a) What is a stationary time series? Describe the ARMA model. 7
b) How can we be sure that an event A causes an event B? Differentiate 8
correlation and causation with an appropriate example.
- 7 Write short notes on: (Any two) 2*5=10
a) Granger Causality
b) Statistical significance
c) The KDD process

*****Best Wishes*****